

Junyuan Pang

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Education

Zhejiang University	School of Mathematical Sciences, B.S. in Mathematics and Applied Mathematics	2020.09 – 2024.06
- GPA: 3.7/4.0 (Rank: 6/20), Strengthening Foundation Program (CKC) - Honors: CKC Navigation Scholarship, CKC Discipline Excellence Scholarship, Zhejiang University Third Prize Scholarship - Relevant Courses: Analysis I (95), Algebra I (95), Geometry (90), Algebra II (94), Topology (92), Data Structures and Algorithms (92), Numerical Algebra (91), Numerical Analysis (90), Mathematical Software (100)		
Zhejiang University	College of Computer Science and Technology, Ph.D. Student in Computer Science	2024.09 – 2028(expected)
Shenzhen Loop Area Institute	Joint Ph.D. Training Program in Computer Science	2025.09 – 2028(expected)

Competition Experience

2017 National High School Mathematics Olympiad (Provincial First Prize, 28th in the province)	2017.09	
2018 National High School Mathematics Olympiad (Provincial First Prize, 16th in the province)	2018.09	
2019 National High School Mathematics Olympiad (Provincial First Prize, 18th in the province)	2019.09	
12th MathorCup University Mathematical Modeling Challenge (National Second Prize)	Team Member	2022.04
- Developed an optimization model for AGV scheduling in unmanned warehouses; led model formulation, simplification, and paper writing. ✓ Translated real-world operating constraints into a tractable optimization model. ✓ Applied randomized algorithms to generate near-optimal schedules for the provided AGV data. ✓ Incorporated collision-avoidance and deadlock-prevention constraints into the scheduling model.		
2022 Zhejiang University Mathematical Modeling Competition (University First Prize)	Team Member	2022.05
- Developed a routing and placement model for robotic-arm handling of medicinal materials; led model formulation, code development, and the mathematical analysis and proofs in the paper. ✓ Implemented backtracking, bitmask dynamic programming, and ant colony optimization for the underlying traveling salesperson problem. ✓ Applied simulated annealing, genetic algorithms, and ant colony optimization to obtain near-optimal material layouts under different robotic-arm strategies. ✓ Proved that the problem remains NP-hard under edge-weight-threshold and minimum-precision constraints, and that membership of its decision version in NP would imply membership in P and hence $P = NP$, ruling out polynomial-time exact algorithms unless $P = NP$.		
2022 National University Mathematical Modeling Competition (National First Prize)	Team Member	2022.09
- Developed a bearing-only passive-localization model for drone formations; led model formulation, partial code debugging, adjustment-strategy design, and the mathematical analysis and proofs in the paper. ✓ Built a formation-flight and signal-reception model and applied Gaussian fitting and Kolmogorov-Smirnov tests to the provided data. ✓ Derived an exact localization method for circular drone formations when the transmitting drone was unknown, supported by a geometric proof. ✓ Designed a precise localization and adjustment strategy requiring four rounds of signal transmission for circular drone formations. ✓ Applied beam search to general drone formations; simulations achieved the target precision within 20-40 rounds of signal transmission and adjustment.		
2023 National University Mathematical Modeling Competition (Provincial First Prize)	Team Member	2023.09
- Developed a physical model and layout-optimization strategy for a heliostat field; led model formulation, algorithm design, and the mathematical and physical analysis and proofs in the paper. ✓ Modeled sunlight reflection in a three-dimensional heliostat-field coordinate system and calculated key components of optical efficiency. ✓ Formulated heliostat placement as a planar circle-packing problem, improving field-area utilization by up to 43.7%, with typical gains of approximately 35%, and achieving an estimated annual average output of 70 MW. ✓ Characterized how output power varied with mirror height, installation height, and location, and applied grid search to optimize heliostat dimensions and installation height.		

Research Experience

National Key Laboratory of Blockchain and Data Security

Researcher

2023.09 – Present

P-Shapley: Shapley Values on Probabilistic Classifiers

VLDB

Third Author

2023.10 – 2024.08

- Introduced P-Shapley, a valuation metric based on predicted-class probabilities that more precisely distinguishes the contributions of individual training samples.
 - Designed a calibration function to sharpen marginal-contribution differences across individual samples.
 - Provided theoretical analysis and reduced weighted accuracy degradation by 34.1% on Fashion-MNIST and 88.1% on CIFAR-10 relative to TMC-Shapley.

Shapley Value Estimation based on Differential Matrix

SIGMOD

First Author

2023.11 – 2025.06

- Developed a differential-matrix estimator that samples complementary contributions and reconstructs Shapley values via least squares.
 - Leveraged the efficiency axiom to sample complementary-contribution differences rather than Shapley values directly, improving sample efficiency over existing methods.
 - Provided a rigorous theoretical analysis of the estimator's performance.
 - Extended the estimator with Neyman allocation, outperforming Monte Carlo Shapley on the Voting and Iris datasets.

TabularMark: Watermarking Tabular Datasets for Machine Learning

CCS

Third Author

2024.05 – 2024.10

- Developed TabularMark, a hypothesis-testing-based watermarking scheme for numerical and categorical data; embedded random perturbations while preserving utility and detected watermarks with a calibrated one-sample proportion z-test. The project is being extended in collaboration with JPMorgan AI Research and AlgoCRYPT CoE.
 - Designed multi-attribute matching and hypothesis-testing procedures to reduce dependence on data types and primary keys, and established the computational difficulty of watermark removal.
 - Demonstrated detectability, non-intrusiveness, and robustness across real-world regression and classification datasets, and analyzed hyperparameter trade-offs on synthetic data.

ASSS: Adaptive Stratified Sampling for Shapley-like Values

SIGMOD

First Author

2024.06 – 2026.06

- Developed ASSS-A and ASSS-B, budget-adaptive stratified estimators for Shapley, Banzhaf, and Beta-Shapley values that address empty-stratum failures under limited utility-evaluation budgets.
 - Proved unbiasedness and variance reduction for ASSS-A and established high-probability guarantees for ASSS-B; reduced memory from $O(n^2)$ to $O(Ln)$ through adaptive pre-stratification.
 - Reduced Shapley RMSE on Breast Cancer by 66.4% versus GELS and 90.9% versus S-GELS at 600,000 utility evaluations over 20 independent runs.

MIXSV: Mixup with Shapley Values

ICDE

Third Author

2025.03 – Present

- Contributed to MIXSV, a data augmentation method that adjusts Mixup weights using sample Shapley values and uses kNN Shapley for efficient valuation.
 - Analyzed when Shapley-guided weighting tightens distribution-discrepancy bounds and attenuates label noise.
 - Achieved 68.79% accuracy on CIFAR-100 features with 20% label noise, exceeding standard Mixup by 11.10 percentage points.

CROSS: Constructive Sampling for Shapley-like Value Estimation

Draft

First Author

2026.04 – Present

- Designed CROSS, a constructive sampling framework that jointly builds coalition sets under prescribed distributions and fixed utility-evaluation budgets instead of drawing samples independently.
 - Developed global and coalition-size-wise constructions using cyclic structures to balance player and pair coverage across Shapley-like value objectives.
 - Added a pilot-driven allocation scheme that directs the remaining budget toward informative coalition sizes while preserving the target valuation weights.

Student Organizations and Volunteer Activities

Class Representative

2021.03 – 2024.06

- Organized community-building activities, academic exchanges, and offline events; coordinated routine class affairs.

Skills and Other Information

- Programming: C++, Python, MATLAB | Tools: LaTeX, Microsoft Office